



Introduction to predictive modeling in healthcare

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Who are we?

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Predictive Modeling in Healthcare Course

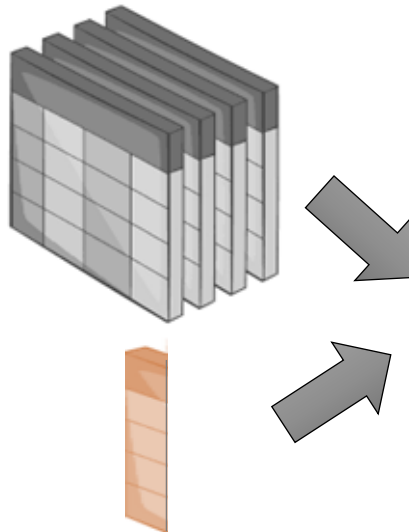
What we will teach

- Bridge the gap between general DS and prediction models in healthcare
- Focus on supervised learning for risk evaluation
- Focus on structured data



Clinical
information:
diagnoses, labs,
hospitalizations,
demographics...

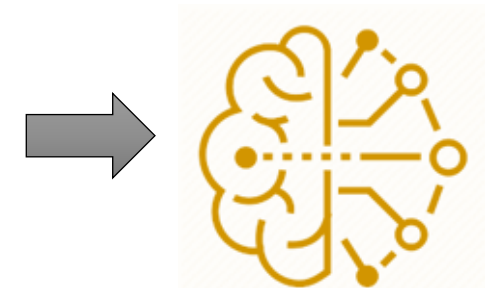
Outcome of
interest



Prediction
Algorithms



A final ML
prediction
model



ML model

What we will teach

- Bridge the gap between general DS and prediction models in healthcare
- Focus on supervised learning for risk evaluation
- Focus on structured data
- Specific topics:
 - Interpretability
 - Fairness
 - Model performance
 - Specific model families, including survival analysis
 - Feature selection
 - Missing data
 - Measuring uncertainty



What we expect from you

- 90% - פרויקט סיכום מודרך במסגרתו יישמו הסטודנטים את החומר שנלמד בקורס תוך פיתוח מודל ניבוי רפואי. תוצרי הפרוייקט יוצגו על ידי הסטודנטים בשיעורים האחרונים לסמסטר.
- 10% - הצגה של פרק מפרוייקט הסיכום (בשיעורים האחרונים של הסמסטר).
- תאריכים (טנטטיביים בלבד, עלולים להתעדכן במהלך הקורס):
 - שיעור Q&A על הפרוייקט:
 - 19/5/2026 (התוכן עד לשיעור שלפני מפגש זה מכסה את התוכן הנדרש לחלקים 1-3 של הפרוייקט)
 - תאריך הגשה של הפרוייקט:
 - תאריך הגשה לחלקים 1-3 של הפרוייקט: 26/5/2026
 - תאריך הגשה לחלקים 4-6 של הפרוייקט: 09/6/2026
 - תאריכי הצגה של פרק מתוך הפרוייקט (נגיד לכל קבוצה איזה פרק להציג, יתכן שההצגות יבוצעו בזום):
 - תאריך הצגה לחלקים 1-3 של הפרוייקט: 16/6/2026 (עד ל- 9/6/2026 נודיע לקבוצות שמציגות בתאריך הזה, כולל מספר החלק שהן מציגות)
 - תאריך הצגה לחלקים 4-6 של הפרוייקט: 23/6/2026 (עד ל- 16/6/2026 נודיע לקבוצות שמציגות בתאריך הזה, כולל מספר החלק שהן מציגות)
- התאמות למילואים (קבוצות 2+3 במתווה) – 100% הערכה מבוססת על פרוייקט ההגשה + גישה להקלטות של השיעורים (רק למילואימניקים)
- מצגות – יועלו בשבוע שבו הועבר השיעור

Relevant bibliography

- Steyerberg, E. W. (2019). Clinical prediction models. Cham: Springer International Publishing.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning (Vol. 112, p. 18). New York: springer. Available at <https://www.statlearning.com/>
- Molnar, C. “Interpretable machine learning. A Guide for Making Black Box Models Explainable”, 2019.
<https://christophm.github.io/interpretable-ml-book/>
- Wickham H. & Grolemund G. (2017). R for Data Science. O’Reilley. Available at <https://r4ds.had.co.nz/>



Course Summary (Rolling) Project



General Intro (1)

- The purpose of this project is mimicking the “real-life” experience of developing a prediction model for healthcare purposes
- We will practice the development of a mortality predictor for intensive care units (ICUs)
- The data we will use will be taken from [this](#) Kaggle challenge
- Potential uses of ICU mortality predictors:
 - Decide who needs to be admitted to the ICU
 - Assess the extent of the intervention measures that should be used
 - Prepare families towards end-of-life decisions

General Intro (2)

- We will not evaluate your code, but you will be asked to submit it (you can use whatever coding language you prefer)
- We will ask to see and evaluate how you present your development outputs, as you would have done for your business client (the head of the ICU in this case)
- The various outputs will reflect the subjects we will touch upon during the course
- This PPT presentation will serve as a template that you will gradually fill with your project outputs during the course
- You can work in groups of 5 students
- In the final lessons, we will ask different groups to present various parts of your project to the class

First meeting with the
head of the ICU

To Do

- This is your first meeting with the Dr. Jane Smith, the head of the ICU
- She asked one of the hospital's analysts to give you access to the retrospective EHR data in her unit, and she is asking you what you can do with it
- This is your opportunity to:
 - Explain a bit about the prediction model you will develop for her
 - Explain what this model can be used for (and what it can't be used for)
 - Show general descriptives of the data set in order to make sure she finds it reliable

Second meeting: show
first modeling results

To Do

- This is your second meeting with the Dr. Jane Smith, the head of the ICU
- Show her your first modeling results. Present 2-3 modeling alternatives (preferably from different model families).
- Prepare slides that describe the general idea of each model you chose to implement
- Explain the pros and cons of deciding to implement each of these model alternatives
- Show the performance of each model (use graphs and tables)
- Show potential intervention cutoffs, and the practical applications of choosing each cutoff

Third meeting:
explainability

To Do

- Use various methods that were presented to give a good overview of model explainability:
 - Present several global explainability alternative
 - Present several local explainability alternative
- Compare the variables in your model compared to existing model alternatives such as the APACHE score (its outputs are in the data)

Fourth meeting:
missing data

To Do

- Explain how you dealt with missing data
- Present how the performance is affected from different alternatives:
 - Complete case analysis
 - Keeping the missing values and using models that can use them as inputs
 - Impute with a “missing” category
 - Various imputing alternatives (at least one of these should be multiple imputation)
- Give examples to specific values that were imputed so that Dr. Smith can give you some input if they seem medically reasonable

Fifth meeting:
describe the uncertainty
in your results

To Do

- Add uncertainty measures to the global performance measures using bootstraps
- Explain the meaning of the uncertainty measures in terms that Dr. Smith can understand

Sixth meeting:
discuss potential unfairness
in the model and how you
addressed it

To Do

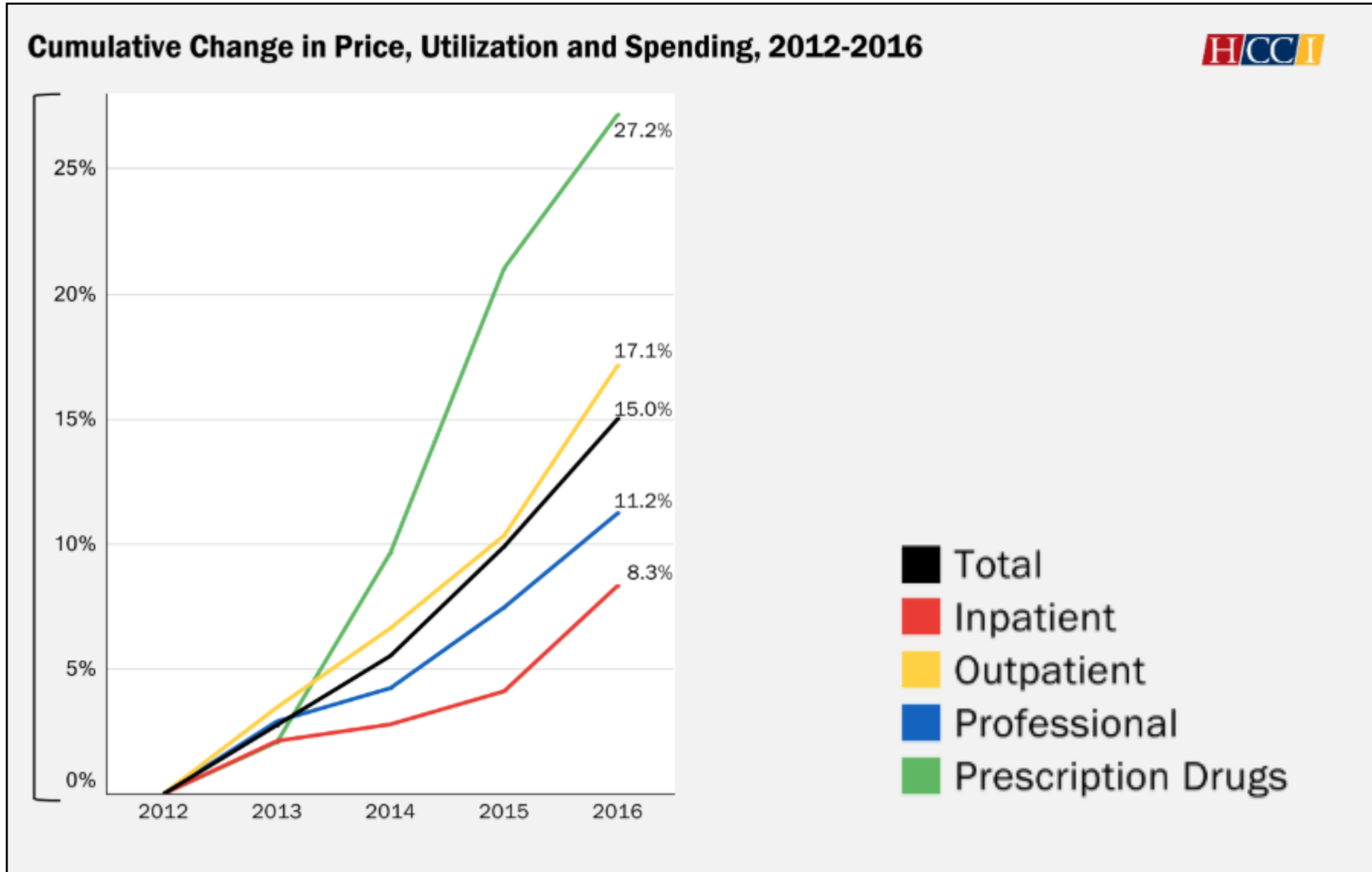
- Explain general considerations of potential fairness issues with regards to the mortality predictor (try to present it in a tangible manner that could be perceived and understood by lay people)
- Consider the following protected variables: age, gender, ethnicity, which hospital the patient was admitted to (potentially proxy of SES?)
- Present performance measures in various subgroups (if you can, try to present the performance for groups of intersecting variables)
- Suggest approaches to dealing with potential unfairness (if you can, try to implement some of them)

Good Luck!

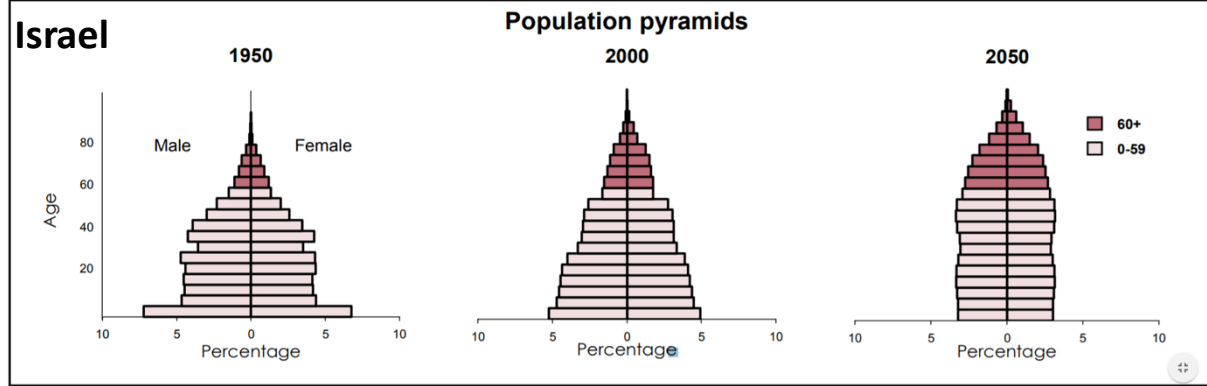
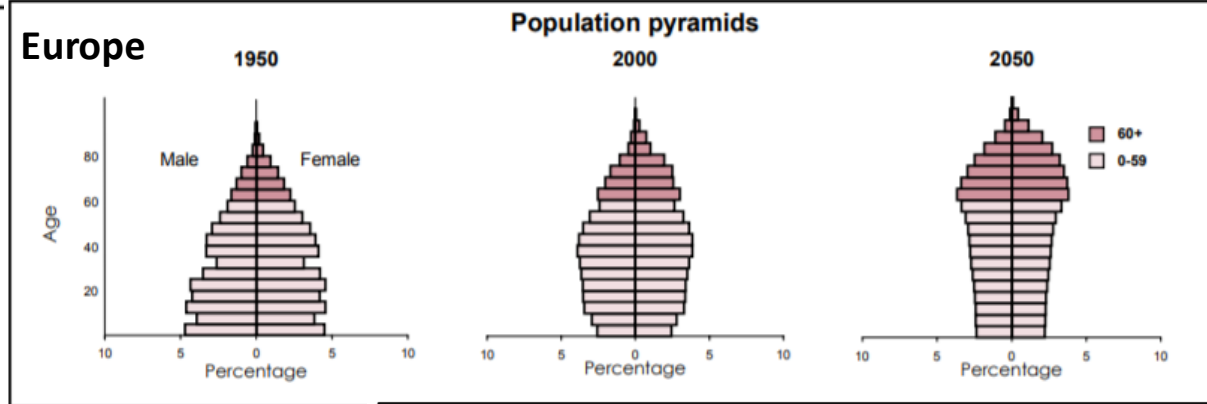
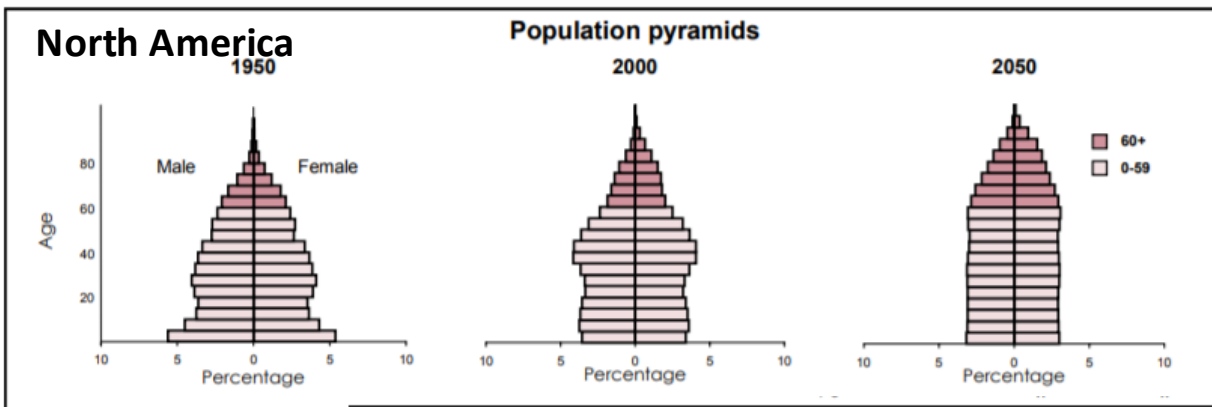


Background

We have a problem...

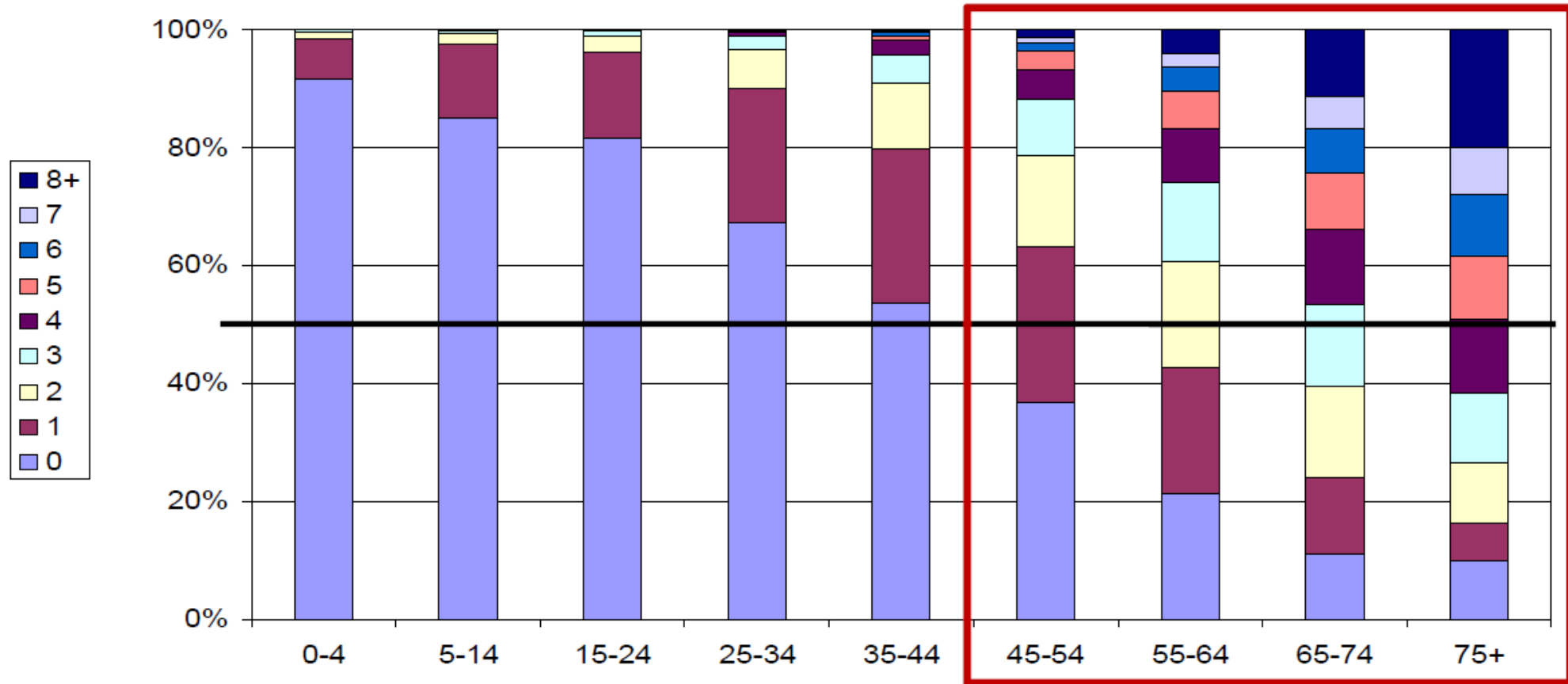


We have a problem...



We have a problem...

Chronic conditions by age



Source: Clalit Research Institute

45%
of necessary
interventions
are missed

- Only 66% of patients with established cardiovascular disease receive statins to lower cholesterol
- Only 25-50% of hypertensive patients achieve appropriate blood pressure control
- Only 30% of those with history of osteoporotic fractures are receiving treatments to prevent a subsequent fracture



We have a problem...

How much time does it take for GPs to provide guideline-recommended preventive, chronic disease, and acute care?

26.7
Hours/day!

[illegible]

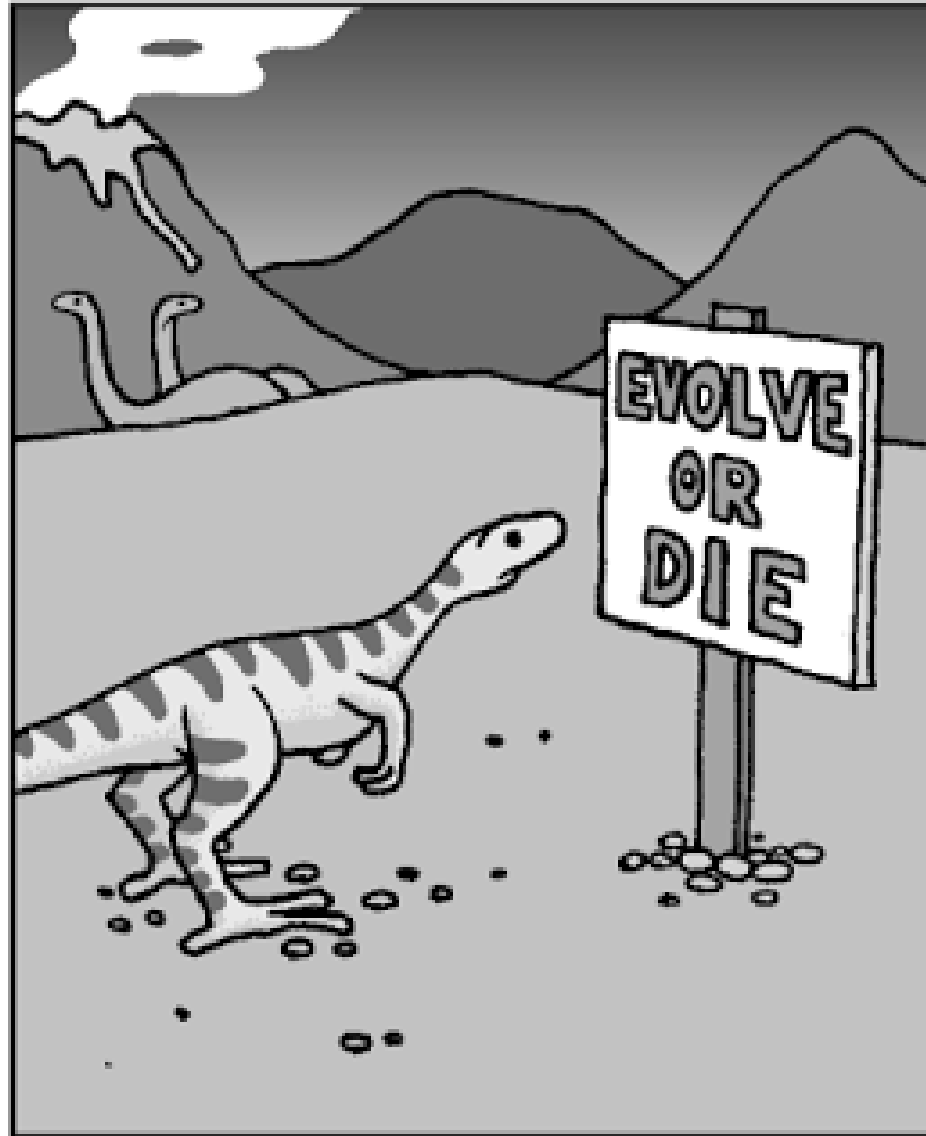
Original Research | Published: 01 July 2022

Revisiting the Time Needed to Provide Adult Primary Care

[Justin Porter MD](#) , [Cynthia Boyd MD, MPH](#), [M. Reza Skandari PhD](#) & [Neda Laiteerapong MD, MS](#)

Journal of General Internal Medicine (2022) | Cite this article

We have a problem...



We have a solution!



Smart use of data

= a requisite and driving force for transforming care

The Israeli Healthcare System

- Since 1995, mandatory health insurance
- A pre-defined “services basket”, including all facets of care, to which everyone is eligible
- Payment and eligibility not tied to one another
- Provided through 4 payer-provider healthcare organizations (“Sick funds”)
- Individuals can change freely – providers can’t reject
- Few do change, around 1-2% per year

The Health Organizations

- Not equal in size. Clalit ~52%, Maccabi ~25%, and two smaller ones
- In general, directly provide
 - Primary care
 - Ambulatory specialist care
 - Labs
 - Imaging
 - Drugs
- Indirectly provide
 - Hospitalizations

Health Data

- Early (~20 years ago) digitalization of electronic health records leading to long-term follow-up
- Each health organization has a unified analytic data warehouse, spanning all facets of an individual's health
- Each person with a single unique identifier
- Interfaces with governmental information systems allows full demographic information



**Know your
limitations**



Current Age ⓘ *

Age must be between 20-79

Sex *

Male

Female

Race *

White

African American

Other

Systolic Blood Pressure (mm Hg) *

Value must be between 90-200

Diastolic Blood Pressure (mm Hg) *

Value must be between 60-130

Total Cholesterol (mg/dL) *

Value must be between 130 - 320

HDL Cholesterol (mg/dL) *

Value must be between 20 - 100

LDL Cholesterol (mg/dL) ⓘ ○

Value must be between 30-300

History of Diabetes? *

Yes

No

Smoker? ⓘ *

Current ⓘ

Former ⓘ

Never ⓘ

On Hypertension Treatment? *

Yes

No

On a Statin? ⓘ ○

Yes

No

On Aspirin Therapy? ⓘ ○

Yes

No



26.9%
High

Current 10-Year
ASCVD Risk**

Lifetime ASCVD Risk: **69%**

Optimal ASCVD Risk: **3.2%**

Current Age ⓘ *

54

Age must be between 20-79

Should I focus on
managing my HTN?



Systolic Blood Pressure (mm Hg) *

145

Value must be between 90-200

Diastolic Blood Pressure (mm Hg) *

90

Value must be between 60-130

Total Cholesterol (mg/dL) *

212

Value must be between 130 - 320

HDL Cholesterol (mg/dL) *

42

Value must be between 20 - 100

LDL Cholesterol (mg/dL) ⓘ ○

135

Value must be between 30-300

History of Diabetes? *

✓ Yes

No

Smoker? ⓘ *

✓ Current ⓘ

Former ⓘ

Never ⓘ

On Hypertension Treatment? *

Yes

✓ No

On a Statin? ⓘ ○

Yes

✓ No

On Aspirin Therapy? ⓘ ○

Yes

✓ No



↓ **18.8%**
Intermediate

Current 10-Year
ASCVD Risk**

Lifetime ASCVD Risk: **69%**

Optimal ASCVD Risk: **3.2%**

Current Age ⓘ *

54

Age must be between 20-79

Sex *

✓ Male

Female

Race *

✓ White

African American

Other

Systolic Blood Pressure (mm Hg) *

115

Value must be between 90-200

Diastolic Blood Pressure (mm Hg) *

80

Value must be between 60-130

Total Cholesterol (mg/dL) *

212

Value must be between 130 - 320

HDL Cholesterol (mg/dL) *

42

Value must be between 20 - 100

LDL Cholesterol (mg/dL) ⓘ ○

135

Value must be between 30-300

History of Diabetes? *

✓ Yes

No

Smoker? ⓘ *

✓ Current ⓘ

Former ⓘ

Never ⓘ

On Hypertension Treatment? *

Yes

✓ No

On a Statin? ⓘ ○

Yes

✓ No

On Aspirin Therapy? ⓘ ○

Yes

✓ No



26.9%
High

Current 10-Year
ASCVD Risk**

Lifetime ASCVD Risk: **69%** Optimal ASCVD Risk: **3.2%**

Current Age ⓘ *

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Age must be between 20-79

Sex *

✓ Male

Female

Race *

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LDL Cholesterol (mg/dL) ⓘ ○

135

Value must be between 30-300

History of Diabetes? *

✓ Yes

No

Former ⓘ

Never ⓘ

On Hypertension Treatment? *

Yes

✓ No

On a Statin? ⓘ ○

Yes

✓ No

On Aspirin Therapy? ⓘ ○

Yes

✓ No

What if I start
taking BP
medication?





30.9%
High

Current 10-Year
ASCVD Risk**

Lifetime ASCVD Risk: **69%** Optimal ASCVD Risk: **3.2%**

Current Age ⓘ *

54

Age must be between 20-79

Sex *

✓ Male

Female

Race *

✓ White

African American

Other

Systolic Blood Pressure (mm Hg) *

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135

Value must be between 30-300

History of Diabetes? *

✓ Yes

No

Former ⓘ

Never ⓘ

On Hypertension Treatment? * ○

✓ Yes

No

On a Statin? ⓘ ○

Yes

✓ No

On Aspirin Therapy? ⓘ ○

Yes

✓ No

What if I start
taking BP
medication?



AI / ML model



The data scientist's tool kit in digital healthcare

Descriptive

Understand your data or **describe what happened** through **descriptive statistics**



Predictive

Identify personalized risks, perform prioritization or predict **what will happen** using **prediction models**



Prescriptive

Evaluate potential interventions to determine **what we should** do through **causal inference**

